

## Research Interests

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- **Multi-modal Representation Learning:** unified modeling of language, vision, and audio in large-scale foundation models.
- **Embodied AI Systems:** 3D perception and domain adaptation for real-world embodied agents.

## 🏛️ Education

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### Korea University

Ph.D. in Artificial Intelligence

GPA: 4.28 / 4.5

Advisor: [Prof. Sangpil Kim](#)

*Seoul, South Korea*

Sep. 2022 – Aug. 2027 (Expected)

### Inha University

B.S. in Computer Science

GPA: 4.0 / 4.5

Advisor: [Prof. Jinman Jung](#)

*Incheon, South Korea*

Mar. 2016 – Aug. 2022

## 🧰 Research Experience

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### University of Michigan

Visiting Researcher

Advisors: [Prof. Honglak Lee](#)

Projects: World Model, Vision-Language-Action Model

*Ann Arbor, MI, USA*

Aug. 2026 – Aug. 2027 (Expected)

### Samsung Advanced Institute of Technology (SAIT)

Visiting Researcher, Computer Vision Lab

Advisors: [Ph.D. Sujin Jang](#)

Projects: Autonomous Driving, Vision-Language Navigation

*Suwon, South Korea*

Jan. 2024 – Feb. 2025

## 📖 Publications († Equal contribution)

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### Conference Papers

#### [C9] Motion Cues from Image-based Point Tracking for LiDAR Scene Flow Estimation

Gyeongrok Oh<sup>†</sup>, Youngdong Jang<sup>†</sup>, Jong Wook Kim, Hyunju Ryu, Hyung-gun Chi, SeungHyeon Kim, Seungryong Kim, Jonghyun Choi, Sangpil Kim

*Submission, 2026* [\[paper\]](#)

#### [C8] Training-Free Global Geometric Association for 4D LiDAR Panoptic Segmentation

Gyeongrok Oh, Youngdong Jang, Jonghyun Choi, Suk-Ju Kang, Guang Lin, Sangpil Kim

*Submission, 2026* [\[paper\]](#)

#### [C7] Active Test-time Vision-Language Navigation

Heeju Ko, Sungjune Kim, Gyeongrok Oh, Jeongyoon Yoon, Honglak Lee, Sujin Jang, Seungryong Kim, Sangpil Kim

*NeurIPS, 2025* [\[paper\]](#)

#### [C6] Test-time Adaptation for Online Vision-Language Navigation with Feedback-based Reinforcement Learning

Gyeongrok Oh<sup>†</sup>, Sungjune Kim<sup>†</sup>, Heeju Ko, Daehyun Ji, Dongwook Lee, Byung-Jun Lee, Sujin Jang\*, Sangpil Kim\*

*ICML, 2025* [\[paper\]](#)

#### [C5] 3D Occupancy Prediction with Low-Resolution Queries via Prototype-aware View Transformation

Gyeongrok Oh<sup>†</sup>, Sungjune Kim<sup>†</sup>, Heeju Ko, Hyung-gun Chi, Jinkyu Kim, Dongwook Lee, Daehyun Ji, Sungjoon Choi, Sujin Jang\*, Sangpil Kim\*

*CVPR, 2025* [\[paper\]](#)

#### [C4] MEVG: Multi-event Video Generation with Text-to-Video Models

Gyeongrok Oh, Jaehwan Jeong, Sieun Kim, Wonmin Byeon, Jinkyu Kim, Sungwoong Kim, Sangpil Kim

*ECCV, 2024* [\[paper\]](#)

[C3] **CMDA: Cross-Modal and Domain Adversarial Adaptation for LiDAR-based 3D Object Detection**  
Gyusam Chang†, Wonseok Roh†, Sujin Jang, Dongwook Lee, Daehyun Ji, **Gyeongrok Oh**, Jinsun Park, Jinkyu Kim\*, Sangpil Kim\*  
*AAAI, 2024* [paper]

[C2] **Functional Hand Type Prior for 3D Hand Pose Estimation and Action Recognition from Egocentric View Monocular Videos**  
Wonseok Roh, Seung hyun Lee, Won Jeong Ryoo, **Gyeongrok Oh**, Jakyung Lee, Soo Yeon Hwang, Hyung-gun Chi, Sangpil Kim  
*BMVC (Oral), 2023* [paper]

[C1] **Sound-Guided Semantic Video Generation**  
Seung hyun Lee, **Gyeongrok Oh**, Wonmin Byeon, Chanyoung Kim, Won Jeong Ryoo, Sang Ho Yoon, Jihyun Bae, Jinkyu Kim\*, Sangpil Kim\*  
*ECCV, 2022* [paper]

## Journal Papers

[J4] **LVMark: Robust Watermark for Latent Video Diffusion Models**  
Youngdong Jang\*, MinHyuk Jang\*, JaeHyeok Lee, Feng Yang, **Gyeongrok Oh**, Jongheon Jeong, Sangpil Kim  
*IEEE Transactions on Information Forensics & Security, 2026. IF: 8.0* [paper]

[J3] **FPANet: Frequency-based Video Demoireing using Frame-level Post Alignment**  
**Gyeongrok Oh**, Sungjune Kim, Heon Gu, Sang Ho Yoon, Jinkyu Kim, Sangpil Kim  
*Neural Networks, 2025. IF: 6.3* [paper]

[J2] **Robust Sound-Guided Image Manipulation**  
Seunghyun Lee†, Hyung-gun Chi†, **Gyeongrok Oh**, Wonmin Byeon, Sang Ho Yoon, Hyunje Park, Wonjun Cho, Jinkyu Kim\*, Sangpil Kim\*  
*Neural Networks, 2024. IF: 6.3* [paper]

[J1] **Audio-Guided Implicit Neural Representation for Local Image Stylization**  
Seung Hyun Lee†, Sieun Kim†, Wonmin Byeon, **Gyeongrok Oh**, Sumin In, Hyeongcheol Park, Sang Ho Yoon, Sung-hee Hong, Jinkyu Kim\*, Sangpil Kim\*  
*Computational Visual Media, 2024. IF: 18.3* [paper]

## 💡 Patents

[P1] **Method and Apparatus with 3D Occupancy Prediction Learning** US Patent  
**Gyeongrok Oh**, Sujin Jang, Sangpil Kim, Sungjune Kim, Jinkyu Kim, Dongwook Lee, Dae Hyun Ji  
US20250356594A1, 2025 (Pending)

## 👤 Teaching Experience

**Korea University** Seoul, South Korea  
Teaching Assistant

- Statistics (2023), Computer Vision (2025), Intelligent Mobility (2025)

**LG CNS** Seoul, South Korea  
Teaching Assistant

- Anomaly Detection Course for Executives and Employees (2022, 2023)

**Inha University** Incheon, South Korea  
Teaching Assistant

- Data Structures (2021)

## 🏆 Awards & Honors

**BK21 Academic Excellence Scholarship** Korea University

**Departmental Scholarship** Inha University  
Department of Computer Science 2020–2022

## Academic Services

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**Reviewer:** CVPR (2024–2026), ICCV (2025), ECCV (2026), NeurIPS (2026), WACV (2023, 2025–2026), BMVC (2026)

**Program Committee:** AAAI (2025–2026)